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Banking Panics: The Role of the First-Come, First-Served Rule and Information Externalities

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This paper proposes that both the first-come, first-served rule and information externalities are important in causing contagious bank runs. The first-come, first-served rule creates a negative payoff externality among depositors. This payoff externality forces depositors to respond to early noisy information such as failures of other banks. Therefore, failures of a few banks may trigger runs on other banks. Contagious runs may occur even if (i) depositors choose the Pareto-dominant equilibrium when there are multiple equilibria and (ii) the deposit contract is chosen to maximize depositor welfare. The feasibility of reforming the FDIC to impose market discipline is also investigated.

I. Introduction

Moral hazard is always an important factor contributing to banking crises. The savings and loan crisis of the United States demonstrated that distressed financial institutions with negative equity values have a strong incentive to pursue high risks. Imprudent bank lending is also cited as a major cause of banking problems happening to Mex-

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ico, Japan, and Southeast Asian countries in the 1990s. Moreover, the huge losses some banks suffer from their positions in financial derivatives show that there are highly risky investment opportunities that banks can gamble on. These facts imply that the threat of moral hazard is real and serious in the banking industry.

One way to alleviate moral hazard problems in the banking system is to impose market discipline, that is, to have depositors or other creditors of banks share the responsibility of monitoring banks. For example, if depositors have to lose part of their deposits when their banks fail, they will withdraw from banks pursuing unsound risks. The threat of a bank run will force banks to reduce their portfolio risk. However, whether market discipline can efficiently control moral hazard problems is still under debate. The experiences of the United States during the national banking era show that bank runs are contagious. Runs on a few banks may trigger runs on other banks, and a large scale of contagious runs may even result in the collapse of the whole financial system. If the contagious feature cannot be removed from bank runs, then requiring depositors to monitor banks may incur huge social costs. A better understanding of the nature of bank runs can help regulators evaluate various proposals for enforcing market discipline.

In this paper, I set up a model to demonstrate how failures of a few banks may cause runs on other banks. The intuition of the model is as follows. At a bank, some depositors may be better informed than others about the value of bank assets. The informed depositors enjoy an advantage over the uninformed by being able to withdraw earlier in bad states in which the bank cannot fully repay all depositors. Facing this informational disadvantage, the uninformed depositors have an incentive to respond to other sources of information before the value of bank assets is revealed. Failures of other banks can be one such information source. If banks' returns are positively correlated, a high bank failure rate implies that returns of the remaining banks are likely to be low. Although information contained in bank failures is very noisy, the uninformed depositors may still respond to it and withdraw. Knowing that the uninformed depositors will withdraw early, the informed are forced to withdraw early too even if more precise information will soon become available. Therefore, bank runs can be contagious. That is, failures of a few banks may trigger runs on other banks. In this paper, I shall call contagious runs ''panic'' for two reasons. First, during contagious runs, depositors are compelled to respond to early information. Their behavior looks as though they are panicking. Second, because a contagious run is based on noisier information, it incurs higher social costs than a run triggered by more precise information about banks.

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My paper is closely related to several papers in the literature. Diamond and Dybvig (1983) show how the first-come, first-served rule imposed in the deposit contract can cause a bank run to happen. In their model, the sequential service constraint creates a negative payoff externality[1] among depositors and results in two equilibria. A bank run is the Pareto-dominated one. My paper shares a common feature with Diamond and Dybvig in that the payoff externality is important in causing inefficient bank runs. In contrast to the model in Diamond and Dybvig, the results in my model do not rely on the multiple equilibria scenario. It is shown that contagious runs occur even if depositors choose the Pareto-dominant equilibrium when there are multiple equilibria. What I emphasize in this paper is how the negative payoff externality affects the way depositors respond to information, not how it creates multiple equilibria.

Chari and Jagannathan (1988) provide an information-based story to explain bank panics. In their model, a panic run is the phenomenon that uninformed depositors misinterpret liquidity withdrawal shocks as withdrawals caused by pessimistic information about bank assets. In my paper, bank runs are also triggered by information, but the mechanism by which information causes panic is different from theirs. While Chari and Jagannathan define panic as the ex post mistake depositors make during their information updating process, I explain panic as the result that depositors respond to early noisy information because of the payoff externality imposed in the deposit contract.

In terms of whether bank runs are an efficient mechanism for monitoring banks, Calomiris and Kahn (1991) argue that a debt contract together with the sequential service constraint is an optimal banking arrangement.[2] My model reaches a conclusion different from theirs. In their model, disciplining banks by bank runs is efficient because runs are always triggered by precise information about bank assets. I show that when failures of other banks can be observed, the uninformed depositors may start a run before more precise information arrives, so bank runs can be inefficient in disciplining banks.

In the way contagious runs happen, my model is similar to the cascade model in Bikhchandani, Hirshleifer, and Welch (1992). In both papers, some agents' actions generate an information exter-

[1] The negative payoff externality refers to the phenomenon that the withdrawal of a depositor reduces the expected payoff for depositors who have not yet withdrawn. In Diamond and Dybvig, given the optimal deposit contract, some depositors will lose their deposits if all depositors try to withdraw early. Therefore, there is a negative payoff externality among depositors in their model.

[2] In their model, the informed depositors who observe pessimistic information start a run, which results in liquidation of the bank. The run has the merit of controlling the bank manager's moral hazard problem.

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nality, inducing other agents to discard their own information and follow suit.[3] In addition to an information externality, my model also includes the negative payoff externality, which is not discussed in Bikhchandani et al. Complementing the cascade model, my paper provides another example that results in herd behavior.

This model can be used to evaluate proposals for reforming the Federal Deposit Insurance Corporation (FDIC). In this paper, I propose a deposit insurance system that can eliminate panic runs and can induce depositors to respond to more precise information. This system fully protects uninformed depositors and induces informed depositors to monitor banks. Moreover, in this system the informed depositors' deposits are not insured, and their claims are junior to the claim of the deposit insurer. This system shares several common features with the subordinated-debt requirements proposed by Calomiris (1997).[4] Both proposals suggest that banks should be disciplined by informed parties and the informed parties' claims should be junior. The results in this paper imply that subordinated debt can be an efficient mechanism for disciplining banks.

The rest of the paper is organized as follows. Section II presents the model. Section III shows the equilibrium and demonstrates the importance of the negative payoff externality and sequential information arrival. The optimal deposit contract is discussed in Section IV. A proposal for reforming the FDIC is studied in Section V. Section VI contains discussions and concluding remarks.

II. A Model of Contagious Bank Runs

This is a three-date model (dates 0, 1, and 2). There are N banks and numerous depositors. Each bank is owned and managed by a risk-neutral banker. Banks do not have capital; they collect deposits from depositors to make investments. Depositors are atomistic. Each depositor receives an endowment of one dollar at date 0. A depositor can either deposit her endowment at a bank or costlessly store the endowment.[5] Depositors face liquidity shocks. A fraction α of depositors die at date 1, and the remaining depositors die at date 2. As in Diamond and Dybvig (1983), I shall call depositors who die at date 1 early diers and depositors who die at date 2 late diers. Early diers have to consume at date 1, and late diers can defer their consumption to date 2. The utility function of a depositor is

[3] In my model, when a panic run occurs, the informed depositors give up the information they will receive in the near future.

[4] I am grateful to an anonymous referee for pointing out the similarities between the subordinated-debt proposal and the deposit insurance system that I propose.

[5] I assume that a depositor cannot deposit at more than one bank. Also, in contrast to Diamond and Dybvig (1983), in my model depositors cannot invest in long-term projects themselves.

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$$U(c_1, c_2) = \begin{cases} u(c_1) & \text{if she dies early} \\ u(c_1 + c_2) & \text{if she dies late,} \end{cases}$$

where c_1 and c_2 are, respectively, the nonnegative consumption levels of the depositor at dates 1 and 2, and u is a utility function with $u' > 0 > u''$. By the strict concavity of u, depositors are risk-averse. Liquidity shocks are realized at date 1. At date 0, a depositor does not know whether she will become an early dier or a late dier.

At date 0, if depositors make deposits, then each bank invests the proceeds in a long-term project.[6] The investment is divisible and will mature at date 2. Different banks choose different projects, but the expected returns on all long-term projects are the same. For each project, the date 2 cash flow generated by one dollar invested at date 0 is either R or r, $R > 1 > r \geq 0$. I shall call R the good outcome and r the bad outcome. For each long-term project, the probability of a bad outcome is a, and the probability of a good outcome is $1 - a$, where a is a random variable whose value depends on the prospects of the banking industry. Assume that a is equal to a_g if the prospects of the banking industry are favorable and is equal to a_b if the prospects are unfavorable ($1 > a_b > a_g > 0$). Given the realized value of a, the outcomes of two different long-term projects are independent. Let η_0 denote the prior probability of unfavorable banking industry prospects, and let a_0 denote the expected value of a at date 0. We thus have

$$a_0 \equiv \eta_0 a_b + (1 - \eta_0) a_g.$$

Although long-term projects mature at date 2, they can be liquidated earlier. Assume that, whatever the date 2 outcome would be, early liquidation releases one dollar for each dollar invested at date 0.

After a bank invests in a long-term project at date 0, at date 1 the banker receives perfect information about the outcome of the investment. There exists a moral hazard problem between bankers and depositors. At date 1, a banker can liquidate the long-term project and invest in an inefficient short-term project. For the short-term project, the investment of one dollar generates either $\bar{R}$ or nothing at date 2, $\bar{R} > R$. The probability of the outcome $\bar{R}$ is ϵ and $\epsilon\bar{R} < 1$, which implies that the expected return on the short-term project is negative. I assume that a banker does not speculate on the inefficient short-term project if he feels indifferent between speculating and not speculating.

[6] For simplicity, the possibility that a bank may diversify its risks by investing in many long-term projects is not considered. If this assumption is relaxed, the main results in this paper still hold as long as banks cannot perfectly diversify risks and returns of different banks' assets are not perfectly correlated.

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At each bank, a fraction β of depositors also receive perfect information about the bank's long-term investment at date 1. These depositors are called informed depositors. Depositors who do not receive information are called uninformed depositors. At date 0, depositors know whether they are informed or not. An informed depositor's chance of becoming an early dier is the same as that of an uninformed depositor. Information about banks and depositors' liquidity shocks are revealed at the same time. This makes it impossible to distinguish informational withdrawals from liquidity withdrawals. Under these assumptions, early diers (both informed and uninformed) always withdraw at date 1 after liquidity shocks are realized. The informed and uninformed late diers may withdraw at either date 1 or date 2.

For most parts of the paper, the deposit contract is assumed to be exogenous. The issue of the optimal deposit contract will be discussed in Section IV. Assume that all banks offer depositors the same deposit contract, which is a debt contract. For each dollar deposited, banks promise to pay a depositor y if she withdraws at date 1 and x if she withdraws at date 2.[7] Withdrawals by depositors are observable to other depositors. When serving customers, banks adopt the first-come, first-served rule, so depositors arriving first can withdraw earlier. If several depositors withdraw at the same time, the order in which they are served is randomly assigned. Convertibility suspension is not allowed: a bank has to be open if it still has money. A bank is liquidated at date 1 if it is out of money. Define

$$\bar{x}(y) \equiv \frac{R - \epsilon\bar{R}}{1 - \epsilon}\frac{1 - \alpha y}{1 - \alpha}. \tag{1}$$

I assume that the deposit contract (y, x) satisfies

$$x = \bar{x}(y) > y > 1. \tag{2}$$

In Section IV, I shall show that under some circumstances the optimal deposit contract maximizing depositor welfare does satisfy condition (2). The condition $x > y$ is necessary for depositors to make deposits at date 0. If $y \geq x$, all depositors will prefer to withdraw at date 1. The condition $y > 1$ is not a trivial assumption. It will be shown that this assumption creates payoff externalities among depositors and causes panic runs.

The function $\bar{x}(y)$ is the largest return a bank can promise to late withdrawers without causing the moral hazard problem in the good state. This can be seen as follows. Suppose that none of the late

[7] Since long-term projects are divisible, at date 1, banks liquidate part of their long-term projects to pay off withdrawing depositors.

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diers withdraw at date 1. If the outcome of the bank's long-term investment is good, then to prevent a banker from speculating, the deposit contract must satisfy

$$(1 - \alpha y)R - (1 - \alpha)x \geq \epsilon[(1 - \alpha y)\bar{R} - (1 - \alpha)x]. \quad (3)$$

The left-hand side of (3) is a bank's profit if the outcome is good and the banker does not speculate on the short-term project; the right-hand side is his expected profit if he speculates. It can be verified that (3) is equivalent to $x \leq \bar{x}(y)$. On the other hand, when the outcome is bad, a banker will speculate on the short-term project if

$$\epsilon[(1 - \alpha y)\bar{R} - (1 - \alpha)x] > 0. \quad (4)$$

It can be shown that (4) is satisfied when $x = \bar{x}(y)$. Therefore, when $x = \bar{x}(y)$ and none of the late diers withdraw at date 1, a banker will speculate on the inefficient short-term project if and only if the outcome of the bank's investment is bad.

Although liquidity shocks and information about the long-term projects of all banks are revealed at date 1, I assume that the timing of revelation is different for different banks. At date 1, N_1 banks are randomly picked, where N_1 is a constant smaller than N. Liquidity shocks and information of these N_1 banks are revealed first; liquidity shocks and information of the remaining $N - N_1$ banks are revealed later. Depositors learn whether their banks are among the first N_1 banks at date 1 before information and liquidity shocks are revealed.

Information revelation may result in bank failures. Let K_1 denote the number of failed banks among the first N_1 banks. Since bank failures are usually public information, I assume that all depositors at the remaining $N - N_1$ banks learn K_1 at the same time. A large K_1 implies that the prospects of the banking industry are likely to be pessimistic. To distinguish the information contained in K_1 from the information that bankers and informed depositors receive, I shall call the latter ''bank-specific information.'' The timetable of the model is shown in table 1. Finally, I assume that $\beta < 0.5$ and that

$$\alpha + (1 - \alpha)\beta < \frac{1}{R}. \quad (5)$$

The assumption that $\beta < 0.5$ means that the proportion of informed depositors is not large. Condition (5) guarantees that all the informed depositors can successfully withdraw at date 1 if a run is triggered by bank-specific information.[8]

[8] When a run is triggered by bank-specific information, the ratio of deposits withdrawn by the informed depositors and early diers to the total date 0 deposits is $[\alpha + (1 - \alpha)\beta]y$. Given the assumptions that $y < R$ and $\alpha + (1 - \alpha)\beta < 1/R$, this

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TABLE 1

TIMETABLE

Date	Events
Date 0	Depositors decide whether to deposit. If they make deposits, banks invest the proceeds in long-term projects
Date 1	(1) Depositors learn whether their banks are among the first N_1 banks. Depositors then decide whether to withdraw (2) Information and liquidity shocks of the first N_1 banks are revealed. Depositors at these banks decide whether to withdraw (3) Depositors at the remaining $N - N_1$ banks observe K_1, which is the number of failures among the first N_1 banks. According to K_1, depositors at the remaining banks decide whether to withdraw (4) Information and liquidity shocks of the remaining $N - N_1$ banks are revealed. If a panic run has not yet occurred, depositors at the remaining banks decide whether to withdraw (5) For each bank that has not been liquidated, the banker decides whether to speculate on an inefficient short-term project
Date 2	Investment projects mature. If a bank is still open, its depositors who have not yet withdrawn withdraw

III. The Equilibrium

The model is solved backward. I shall first study depositors' date 1 withdrawal decisions under the assumption that they have made deposits at date 0. I shall then discuss the conditions under which depositors prefer depositing to self-storing at date 0. Throughout this section, I assume that depositors choose the Pareto-dominant equilibrium when there are multiple equilibria. The reasons for making this assumption will be explained in Section VI. Also, only symmetric, pure-strategy, subgame-perfect Nash equilibria are studied. For simplicity, depositors are assumed not to withdraw when they feel indifferent between withdrawing and not withdrawing.

Suppose that depositors make deposits at date 0 and none of them withdraw after they learn whether their banks are among the first N_1 banks. For each of the first N_1 banks, if the outcome of the bank's long-term investment is good, the informed late diers will not withdraw.[9] Having observed that the fraction of depositors who withdraw early is α, the uninformed late diers will not withdraw either. On the other hand, if the informed late diers learn that the investment

ratio is strictly smaller than one, which implies that all the informed depositors can successfully withdraw at date 1.

[9] For each of the first N_1 banks, when the outcome of the long-term investment is good, obviously "no late diers withdraw at date 1" is a Nash equilibrium and it Pareto-dominates all the other equilibria. Therefore, if depositors choose the Pareto-dominant equilibrium, the informed late diers will not withdraw in this case.

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outcome is bad, they will withdraw immediately.[10] When the uninformed late diers observe that the fraction of depositors who withdraw early exceeds α, they realize that the investment outcome must be bad, and they will withdraw too. The bank fails in this case.

As mentioned, depositors at the remaining $N - N_1$ banks can observe K_1, which is the number of bank failures among the first N_1 banks. Given K_1, depositors at the remaining banks update the probability that the prospects of the banking industry are unfavorable. Information contained in bank failures is especially valuable for the uninformed depositors. Under the assumption that $\alpha + (1 - \alpha)\beta < 1/R$, informed depositors can always withdraw successfully when a run is triggered by bank-specific information. In contrast, the uninformed may be unable to get their money back if they wait until bank-specific information arrives and the outcome of the long-term investment turns out to be bad. Because of this informational disadvantage, uninformed depositors have a strong incentive to respond to K_1.

Let η_1 denote the posterior probability that the prospects of the banking industry are unfavorable given K_1. Term η_1 can be written as

$$\eta_1(K_1) \equiv \frac{\eta_0 a_b^{K_1}(1 - a_b)^{N_1 - K_1}}{(1 - \eta_0) a_g^{K_1}(1 - a_g)^{N_1 - K_1} + \eta_0 a_b^{K_1}(1 - a_b)^{N_1 - K_1}}. \quad (6)$$

From η_1, depositors calculate a_1, the probability of a bad investment outcome given K_1, where

$$a_1(K_1) \equiv \eta_1(K_1) a_b + [1 - \eta_1(K_1)] a_g. \quad (7)$$

Depositors at the remaining banks use a_1 to decide whether to withdraw in response to K_1. The condition under which depositors will respond to K_1 is shown in proposition 1 below. Lemmas 1 and 2 provide the results necessary for the proof of proposition 1. The proof of lemma 2 is in the Appendix. The proof of lemma 1 will be discussed below.

LEMMA 1. For each of the remaining $N - N_1$ banks, when K_1 arrives, "no depositors withdraw in response to K_1" can be supported as an equilibrium if and only if

$$\frac{a_1(K_1)}{1 - a_1(K_1)} \leq z(y, x) \equiv \frac{(1 - \alpha)(1 - \beta)y}{y - 1} \frac{u(x) - u(y)}{u(y) - u(0)}. \quad (8)$$

LEMMA 2. For each of the remaining $N - N_1$ banks, when K_1 ar-

[10] Given a bad investment outcome, the expected utility of an informed late dier for not withdrawing at date 1 is no larger than $u(1)$, which is strictly smaller than $u(y)$.

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rives, if both ''all depositors withdraw in response to K_1'' and ''no depositors withdraw in response to K_1'' are equilibria, the latter always Pareto-dominates the former.

Since only symmetric pure-strategy equilibria are studied, there are only two possible equilibria when K_1 arrives: all depositors withdraw responding to K_1, and no depositors withdraw.[11] As mentioned in the Introduction, I shall call the former a panic run. When $y > 1$, a panic run is always an equilibrium. Lemma 1 states that ''no depositors withdraw in response to K_1'' can be sustained as a Nash equilibrium when the posterior probability of a bad investment outcome is small. Lemma 2 shows that the panic run equilibrium is always Pareto-dominated by the no-withdrawal equilibrium when both are equilibria. Therefore, if we assume that depositors choose the Pareto-dominant equilibrium when there are multiple equilibria, a panic run will occur if and only if (8) is violated.

The intuition behind lemma 1 can be explained as follows. ''No depositors respond to K_1'' can be supported as an equilibrium if, under the belief that no others respond to K_1, no depositors want to withdraw when K_1 is revealed. Since uninformed depositors have a stronger incentive to respond to K_1, the no-withdrawal equilibrium can be sustained if none of the uninformed depositors have the incentive to respond to K_1.

Consider the withdrawal decision of an uninformed depositor who believes that no other depositors at her bank will respond to K_1. If she withdraws in response to K_1, her utility is $u(y)$. On the other hand, if she waits until liquidity shocks and information about her bank are revealed, her expected utility is

$$u_g(y, x) \equiv \alpha u(y) + (1 - \alpha) u(x) \tag{9}$$

if the bank's investment outcome is good and is[12]

[11] Given K_1, there are two other equilibrium candidates. One is that all the informed depositors withdraw and none of the uninformed depositors do. The other is that all the uninformed depositors withdraw and none of the informed depositors do. The former is never an equilibrium because the uninformed depositors have stronger incentives to respond to K_1 than the informed. As to the latter, when $\beta < 0.5$, if it can be sustained as an equilibrium, then it is always Pareto-dominated by the equilibrium in which no depositors withdraw in response to K_1 (the proof of this result is available from the author on request). Therefore, we need not consider this equilibrium candidate when $\beta < 0.5$ and the Pareto-dominance criterion is imposed.

[12] Suppose that the realized outcome of the bank investment is bad. If the depositor becomes an early dier, she can always successfully withdraw, so her utility is $u(y)$. On the other hand, if she becomes a late dier, her expected utility is

$$\frac{1 - [\alpha + (1 - \alpha)\beta]y}{(1 - \alpha)(1 - \beta)y} u(y) + \frac{y - 1}{(1 - \alpha)(1 - \beta)y} u(0).$$

In this expression, $\{1 - [\alpha + (1 - \alpha)\beta]y\}/[(1 - \alpha)(1 - \beta)y]$ is the probability that the uninformed late dier can successfully withdraw at date 1 given the bad outcome.

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$$u_b(y) \equiv \alpha u(y) + (1 - \alpha)\left\{\frac{1 - [\alpha + (1 - \alpha)\beta]y}{(1 - \alpha)(1 - \beta)y}\, u(y) + \frac{y - 1}{(1 - \alpha)(1 - \beta)y}\, u(0)\right\} \tag{10}$$

$$= \frac{1 - \beta y}{(1 - \beta)y} u(y) + \frac{y - 1}{(1 - \beta)y} u(0)$$

if the bank's investment outcome is bad. Given K_1, the depositor's expected utility for not withdrawing is

$$a_1(K_1)\, u_b(y) + [1 - a_1(K_1)]\, u_g(y, x). \tag{11}$$

The uninformed depositor will not respond to K_1 if and only if (11) is no less than $u(y)$. It can be shown that this condition is equivalent to (8) in lemma 1.

Let K_1^* denote the lowest K_1 that will induce all depositors at the remaining $N - N_1$ banks to withdraw early. If $a_g/(1 - a_g) > z(y, x)$, then, from lemma 1, depositors at the remaining banks always respond to K_1 for whatever K_1 is, so $K_1^* = 0$.[13] On the other hand, if $a_b/(1 - a_b) \leq z(y, x)$, from lemma 1, depositors at the remaining banks never respond to K_1, so $K_1^* = N_1 + 1$. If $a_g/(1 - a_g) \leq z(y, x) < a_b/(1 - a_b)$, then using equations (6) and (7) to rearrange condition (8), we can show that K_1^* is the minimum K_1 that satisfies

$$K_1 > \frac{\log\left(\frac{1 - \eta_0}{\eta_0}\right) + \log\left(\frac{1 - a_g}{1 - a_b}\right) + \log\left[\frac{z(y, x) - \frac{a_g}{1 - a_g}}{\frac{a_b}{1 - a_b} - z(y, x)}\right]}{\log\left[\frac{a_b(1 - a_g)}{a_g(1 - a_b)}\right]} + \frac{\log\left(\frac{1 - a_g}{1 - a_b}\right)}{\log\left[\frac{a_b(1 - a_g)}{a_g(1 - a_b)}\right]} N_1. \tag{12}$$

Proposition 1 summarizes depositors' behavior. The proof of proposition 1 is in the Appendix.

[13] Note that $a_1(K_1)/[1 - a_1(K_1)]$ is increasing in $a_1(K_1)$ and $a_g \leq a_1(K_1) \leq a_b$.

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PROPOSITION 1. Assume that depositors choose the Pareto-dominant equilibrium when there are multiple equilibria. Having observed the behavior of depositors at the N_1 banks on which information is revealed early, depositors at the remaining $N - N_1$ banks will start a panic run if and only if $K_1 \geq K_1^*$. Moreover, K_1^* is weakly decreasing in η_0 and y and is weakly increasing in x.

Proposition 1 states that, even if bank-specific information will be available and depositors choose the Pareto-dominant equilibrium, contagious runs can still be triggered by failures of other banks. Both payoff externalities and information externalities are important in causing panic runs. If $y = 1$, there are no payoff externalities among depositors, so depositors always wait until bank-specific information is revealed.[14] On the other hand, if information about all banks arrives at the same time, depositors have no chance to respond to failures of other banks, so contagious runs never happen.

Proposition 1 also shows the conditions under which a panic run is more likely to occur. The probability of a panic run is higher when the prospects of the banking industry are more likely to be unfavorable. That is, K_1^* is decreasing in η_0. This is consistent with the empirical evidence that large-scale bank runs followed pessimistic macroeconomic information (see Gorton 1988; Calomiris and Gorton 1991). Also, panic runs are more likely to occur if either an early withdrawer gets more (the larger the y) or a late withdrawer gets less (the smaller the x). A larger y or a smaller x makes early withdrawals less costly for depositors and thus increases the possibility of a panic run. This result implies that one way to eliminate the threat of panic runs is to adjust the deposit contract. This issue will be discussed in the next section.

At date 1, depositors may have an incentive to withdraw when they learn whether their banks are among the first N_1 banks.[15] To find the condition under which they will not do so, let p_b^p and p_g^p denote the probabilities that a panic run will occur when the prospects of the banking industry are unfavorable and favorable, respectively. If $K_1^* > N_1$, a panic run never occurs, so $p_b^p = p_g^p = 0$. If $K_1^* \leq 0$, a panic run always occurs, so $p_b^p = p_g^p = 1$. If $1 \leq K_1^* \leq N_1$, we have

$$p_b^p \equiv \sum_{i=K_1^*}^{N_1} \binom{N_1}{i} a_b^i (1 - a_b)^{N_1 - i} \tag{13}$$

[14] Note that $z(1, x)$ approaches infinity, which is always larger than $a_b/(1 - a_b)$. So $K_1^* = N_1 + 1$ in this case.

[15] The reason is that at date 1 depositors' expected utilities depend on whether their banks are among the first N_1 banks.

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and

$$p_g^p \equiv \sum_{i=K_1^*}^{N_1} \binom{N_1}{i} a_g^i (1 - a_g)^{N_1 - i}. \tag{14}$$

Using p_b^p and p_g^p, we can calculate depositors' expected utilities. Let $\overline{U}_{u1}$ denote the expected utility of an uninformed depositor at the first N_1 banks and let $\overline{U}_{u2}$ denote the expected utility of an uninformed depositor at the remaining banks.[16] The expressions for $\overline{U}_{u1}$ and $\overline{U}_{u2}$ are in the Appendix. When depositors learn whether their banks are among the first N_1 banks, no depositors will withdraw if and only if[17]

$$\min\{\overline{U}_{u1}, \overline{U}_{u2}\} \geq u(y). \tag{15}$$

Now we go back to date 0, when depositors determine whether to make deposits. Let $\overline{U}_i$ and $\overline{U}_u$ denote the date 0 expected utilities of the informed and uninformed depositors, respectively. The expression for $\overline{U}_i$ is in the Appendix, and $\overline{U}_u$ can be written as

$$\overline{U}_u \equiv \frac{N_1}{N}\overline{U}_{u1} + \frac{N - N_1}{N}\overline{U}_{u2}. \tag{16}$$

Obviously $\overline{U}_i > \overline{U}_u$. Because the informed depositors enjoy an informational advantage over the uninformed, their expected utility from making deposits is higher. From the definition of $\overline{U}_u$ and the assumption that $y > 1$, $\overline{U}_u \geq u(y) > u(1)$ if (15) is satisfied. Therefore, all depositors will choose to deposit at date 0 if (15) holds.[18]

IV. The Optimal Deposit Contract

It has been shown in the previous section that failures of a few banks may trigger panic runs on other banks. Proposition 1 implies that

[16] Note that $\overline{U}_{u1}$ and $\overline{U}_{u2}$ are uninformed depositors' expected utilities after depositors learn whether their banks are among the first N_1 banks and before liquidity shocks and bank-specific information are revealed.

[17] Because the uninformed depositors have a stronger incentive to withdraw than the informed, the informed will not withdraw if the uninformed will not withdraw.

[18] Note that depositors may still deposit at date 0 when (15) is violated. It can be shown that if $\overline{U}_{u1} \leq u(y)$, then depositors never make deposits at date 0. However, if $\overline{U}_{u1} > u(y) > \overline{U}_{u2}$, then depositors will deposit at date 0 if and only if

$$\frac{N_1}{N}\overline{U}_{u1} + \frac{N - N_1}{N}\left[\frac{1}{y}u(y) + \frac{y - 1}{y}u(0)\right] \geq u(1).$$

When this condition holds and $\overline{U}_{u1} > u(y) > \overline{U}_{u2}$, all the depositors at the remaining $N - N_1$ banks withdraw at date 1 once they realize that their banks are not among the first N_1 banks. In this case, a panic run occurs before any information about bank assets is revealed. In the rest of the paper, I shall discuss only the case in which (15) is satisfied.

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one way to eliminate panic runs is to increase x or decrease y. A natural question follows: If the deposit contract can be endogenously determined, will panic runs still occur in equilibrium? It might always be optimal to choose a deposit contract under which panic runs never happen. The scenario in this paper would become less convincing if this were true.

To answer this question, I modify the model. Instead of assuming that the deposit contract is exogenously given, I assume that the deposit contract maximizes depositor welfare, and every depositor has the same weight when welfare is calculated.

Let a_2 denote the probability that a bank will be liquidated at date 1. We have[19]

$$a_2 \equiv \frac{N_1}{N} a_0 + \frac{N - N_1}{N} \{\eta_0 [p_b^p + (1 - p_b^p) a_b] + (1 - \eta_0) [p_g^p + (1 - p_g^p) a_g]\}. \tag{17}$$

The optimization problem is

$$\max_{(y,x)} a_2 \left[\frac{1}{y} u(y) + \left(1 - \frac{1}{y}\right) u(0) \right] + (1 - a_2) u_g(y, x) \tag{18}$$

subject to (15) and

$$\bar{x}(y) \geq x > y \geq 0. \tag{19}$$

The objective function in (18) can be explained as follows. With probability a_2, a bank is liquidated at date 1. In this case, a depositor's utility is $u(y)$ if she withdraws successfully and is $u(0)$ if she loses her deposits. The probabilities of these two events are $1/y$ and $1 - (1/y)$, respectively. With probability $1 - a_2$, a bank is not liquidated at date 1. In this case, the utility of an early dier is $u(y)$ and the utility of a late dier is $u(x)$.[20] Condition (15) guarantees that all depositors prefer to make deposits at date 0 and no depositors

[19] Equation (17) can be explained as follows. If a bank is among the first N_1 banks, then the probability that it will be liquidated at date 1 is a_0. If a bank is among the remaining banks, it will be liquidated at date 1 either because of a panic run or because of an informed run. If the prospects of the banking industry are unfavorable, the probability of a panic run is p_b^p and the probability of an informed run is $(1 - p_b^p) a_b$. Term η_0 is the prior probability of unfavorable banking industry prospects. The case in which the prospects of the banking industry are favorable can be explained in a similar way.

[20] Since informed depositors receive perfect information about their banks, the outcome of a bank's investment must be good if the bank is not liquidated at date 1. Also, because every depositor has the same weight when depositor welfare is calculated, in the objective function there is no need to distinguish between the informed and the uninformed depositors.

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withdraw when they learn whether their banks are among the first N_1 banks. As to condition (19), $x > y$ is necessary since all depositors will prefer to withdraw at date 1 if $y \geq x$; $\bar{x}(y) \geq x$ makes sure that a banker will not speculate on the short-term project when the outcome of the bank's long-term investment is good.

Although a_2 is a function of x and y, it is not differentiable with respect to x and y. This makes solving the optimal (y, x) very difficult. Here I shall briefly discuss the basic trade-offs in determining the optimal deposit contract. Since the contract maximizes depositor welfare, it is obvious that x should be equal to $\bar{x}(y)$.[21] Note that $\bar{x}(y)$ is decreasing in y. To eliminate panic runs, y should be set small and x should be set large so that depositors have more incentive to wait. However, doing so may weaken the merit of the deposit contract in insuring depositors' liquidity risks. As Diamond and Dybvig (1983) show, risk sharing among depositors can be improved if early diers are allowed to consume more.

Intuitively, panic runs would be allowed to happen in equilibrium if (i) risk sharing among depositors is important, (ii) the threat of panic runs is not serious at date 0, and (iii) a large K_1 dramatically raises the posterior probability that the remaining banks are in a bad state.[22] These conditions together reflect a situation in which the major concern for designing the deposit contract is risk sharing rather than the prevention of panic runs, and a large K_1 makes depositors pessimistic about the prospects of the banking industry. Proposition 2 states that numerical examples in which panic runs occur in equilibrium do exist.

PROPOSITION 2. Assume that depositors choose the Pareto-dominant equilibrium when there are multiple equilibria. Even if the deposit contract is designed to maximize depositor welfare, there are cases in which panic runs occur with positive probability in equilibrium.

The following is a numerical example. Assume that $\alpha = 0.3$, $\beta = 0.1$, $R = 1.2$, $\epsilon = 0.1$, $\bar{R} = 1.6$, $a_g = 0.0001$, $a_b = 0.1$, $\eta_0 = 0.005$, $N = 1{,}000$, $N_1 = 1$, and $u(c) = 1 - \exp(-5c)$. The optimal deposit contract (y^*, x^*) is (1.0653, 1.1232) and $K_1^* = 1$. It can be verified that the largest y that will not cause panic runs is 1.02016 (the corre-

[21] If $x < \bar{x}(y)$, an increase in x will raise depositor welfare by reducing the probability of a panic run and by increasing the late diers' payoff when a panic run does not happen.

[22] The first condition is satisfied if the depositors' utility function meets certain requirements. For example, Diamond and Dybvig (1983) require that the measure of relative risk aversion $-cu''(c)/u'(c)$ be larger than one. The second condition is satisfied when both η_0 and a_g are small. The third condition is satisfied when a_b is large.

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sponding x is 1.14557). However, setting y equal to 1.0202 is dominated by setting y equal to 1.0653.

V. Policy Implications for Reforming the FDIC

This paper proposes that bank runs are depositors' rational response to information rather than mysterious panic. However, this claim does not necessarily imply that bank runs can efficiently discipline banks. Because of the negative payoff externalities in the deposit contract, bank runs can be triggered by very noisy information. To enforce market discipline, there should be a mechanism that can induce depositors to be more patient in responding to information.

In this section, I show that there is a deposit insurance system that can eliminate panic runs and can induce depositors to respond to bank-specific information. The basic idea behind this system is simple. Because uninformed depositors have a stronger incentive to start a panic run, they should enjoy more insurance coverage. One way to do so is to fully insure the uninformed depositors and offer no insurance to the informed depositors. The fully protected uninformed depositors need not withdraw early unless they have liquidity needs. The uninsured informed depositors at a bank still have to withdraw early when the outcome of the bank's long-term investment is bad. However, because the uninformed never start the run, the informed depositors can always wait until bank-specific information is revealed. It is shown that this system can be incentive compatible. That is, even if the deposit insurer cannot distinguish between the informed and the uninformed, the uninformed depositors will voluntarily buy deposit insurance, whereas the informed will not.

For simplicity, assume that the insurer is the government. Given a deposit contract described in Section II,[23] the insurer offers the following deposit insurance system. (i) An insured depositor receives y if she withdraws at date 1 and receives $x - \pi$ if she withdraws at date 2, where π is the insurance premium charged by the insurer.[24] (ii) A bank is closed by the insurer at date 1 if the fraction of depositors who withdraw early reaches $\alpha + (1 - \alpha)\beta$. (iii) If a bank is closed by the insurer at date 1, the uninsured depositors who have not yet withdrawn lose all their deposits. In other situations, the uninsured depositors at a bank are paid by the bank according to the deposit contract and the amount of money left at the bank. (iv) If

[23] The purpose of this section is not to find the optimal combination of the deposit contract and the deposit insurance system, but to show how a deposit insurance system may improve welfare given a deposit contract described in Sec. II.

[24] If an insured depositor is not fully paid by her bank, the insurer gives the depositor the difference.

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a bank's realized fraction of depositors who withdraw early is between α and $\alpha + (1 - \alpha)\beta$, the insurer subsidizes the bank so that the banker will not speculate on the short-term project.[25]

The following proposition documents the result that the proposed deposit insurance system can eliminate panic runs. Its proof is in the Appendix.

PROPOSITION 3. Suppose that (i) the deposit contract is (y, x), where $x = \bar{x}(y) > y > 1$, and (ii) without deposit insurance, panic runs occur with positive probability. There exists a deposit insurance system with the following properties. (*a*) In this system, all the uninformed depositors buy deposit insurance and no informed depositors do. (*b*) In this system, panic runs never occur, and a bank is liquidated at date 1 if and only if the outcome of its long-term investment is bad. (*c*) Bankers, informed depositors, and uninformed depositors are better off than in the case without deposit insurance. Moreover, in this system the insurer's expected profit may be positive.

As proposition 3 shows, in the deposit insurance system, panic runs are eliminated. Bank runs are efficient because they are based on bank-specific information. Moreover, it is possible that the welfare of all parties, including depositors, bankers, and the deposit insurer, is improved. The success of the proposed system in enforcing market discipline without triggering panic runs can be attributed to the following features of the system. First, banks are monitored only by their informed depositors. By fully protecting the uninformed depositors, this system allows the informed depositors to be more patient in responding to information. Second, the claims of the uninsured depositors are junior to the claim of the deposit insurer.[26] Because of this feature, an uninformed depositor who does not buy insurance will lose all of her deposits once a run is triggered by bank-specific information.[27] This gives uninformed depositors a strong incentive to buy deposit insurance. Third, the system requires the insurer to subsidize sound banks suffering liquidity problems.

[25] Assume that the subsidy can be given to the bank before the banker decides whether to speculate. It can be shown that the ratio of the required subsidy to the bank's total date 0 deposits is

$$\frac{(y - 1)(R - \epsilon\bar{R})(t - \alpha)}{(1 - \alpha)(1 - \epsilon)},$$

where t is the fraction of depositors who withdraw.

[26] This feature is implied by the requirement that uninsured depositors lose all their deposits when banks are closed by the insurer.

[27] Note that a bank is closed when the fraction of depositors who withdraw early reaches $\alpha + (1 - \alpha)\beta$. When an uninformed depositor finds that the outcome of the bank's investment is poor, it is too late for her to withdraw.

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The subsidy prevents bankers with good investment outcomes from speculating on inefficient short-term projects. Knowing this, an informed depositor need not panic even if she believes that all the other informed depositors will withdraw at date 1. Therefore, there is only one subgame-perfect Nash equilibrium in this system, which implies that it is no longer necessary to assume that depositors choose the Pareto-dominant equilibrium.

In addition to the merits that have been mentioned, the proposed deposit insurance system may have others. For example, consider the case in which information collection is costly and whether a depositor is informed is endogenously determined. In this case, requiring uninsured depositors' claims to be junior gives these depositors a strong incentive to acquire information and monitor banks. Moreover, inducing only uninsured depositors to monitor banks can avoid duplication of efforts expended on information acquisition.

A problem with this deposit insurance system is that it relies on the strong assumption that α and β are constants. If α and β are random variables, the design of the deposit insurance system would become much more complicated. However, the insights provided in the analysis above—that banks should be monitored by informed depositors with junior claims and that the deposit insurer should subsidize sound banks suffering liquidity problems—will be robust to the change of assumptions.

One thing worth mentioning is that the deposit insurance system discussed in this section shares several common features with the subordinated-debt requirements proposed by Calomiris (1997). Calomiris suggests that commercial banks should be required to finance a minimal fraction of their total nonreserve assets with subordinated debt earning a yield no greater than 50 basis points above the riskless rate. Under this proposal, banks have to reduce their portfolio risk so that investors are willing to hold their subordinated debts. Moreover, subordinated-debt holders of a bank have a strong incentive to monitor the bank because they are likely to receive nothing if the bank fails. The role of subordinated-debt holders in Calomiris's proposal is similar to that of informed depositors in the deposit insurance system I propose. Both proposals suggest that banks should be monitored by informed parties and the informed parties' claims should be junior. A policy implication of this paper is that, for enforcing market discipline in the banking industry, the subordinated-debt approach has advantages over the fractional-insurance approach. Under a fractional-insurance reform proposal, deposits are fractionally insured. In this case, all depositors have incentives to acquire information about banks, and all depositors have incentives to withdraw in response to various sources of information. From

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discussions in this section, it should be obvious that both the probability of a panic run and the social costs incurred in monitoring banks will be lower under the subordinated-debt proposal than under a fractional-insurance proposal.

VI. Discussion and Concluding Remarks

In this paper, I set up a model with multiple banks to study the contagious nature of bank runs. The model suggests that both payoff and information externalities are important in causing inefficient runs. This paper also discusses a plausible proposal for reforming the FDIC. I show that there is an incentive-compatible deposit insurance system that can make bank runs an efficient mechanism for disciplining banks.

In most parts of this paper, I assume that depositors choose the Pareto-dominant equilibrium when there are multiple equilibria. The reasons for adopting this Pareto-dominance criterion can be explained as follows. First, the results I get under this assumption are consistent with the empirical evidence that bank runs are triggered by pessimistic information about banks. Second, by making this assumption, I can concentrate on how the negative payoff externality affects depositors' incentives in responding to information rather than how it creates multiple equilibria. Third, one purpose of this paper is to demonstrate that bank runs may not be an efficient mechanism for monitoring banks. If it can be shown that a panic run can occur even if depositors choose the Pareto-dominant equilibrium, panic runs will occur more frequently when depositors may choose a Pareto-dominated equilibrium. Finally, there is evidence that under some circumstances the behavior of economic agents does converge to what is predicted by the Pareto-dominant equilibrium (see, e.g., Harrison and Hirshleifer 1989). I conjecture that most of the results in this paper will hold if the Pareto-dominance criterion is replaced by the assumption that depositors are more likely to withdraw when the posterior probability of a bad investment outcome is higher.

In addition to the equilibrium selection criterion, the assumptions about the information structure also deserve discussion. I assume that bank-specific information always exists, so depositors can always receive better information if they do not respond to failures of other banks. I also assume that no costs are incurred between the information release of the first group of banks and that of the remaining banks. These assumptions imply that depositors should wait until bank-specific information arrives to decide whether to withdraw. If failures of other banks were the only information available to deposi-

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tors or if banks would suffer great losses supposing that depositors did not respond immediately to failures of other banks, then panic runs might be socially efficient. The assumptions in the model are justified by the following arguments. First, during the bank panics in U.S. history, the authorities or clearinghouse associations always played the role of verifying the states of banks. In general, the process of information revelation took about one month. Therefore, if depositors were patient, bank-specific information would be available. Second, according to Gilbert (1990), most empirical studies show that the stock price of an individual bank accurately reflects the risk assumed by the bank, which implies that bank-specific information exists. Third, the span of a panic was always shorter than three months. It is not plausible that large losses could be incurred in such a short time.

There are several directions to extend the present model. One is to assume that information collection is costly and depositors' information acquisition decisions are endogenously determined. Since depositors' incentives to collect information may be different under different reform proposals, this extension can offer new insights into reform of the FDIC. Another direction of extending this paper is to study the case in which the proportions of early diers and informed depositors are random variables rather than constants. The information-updating process of the uninformed depositors will become more complicated. Issues discussed in Chari and Jagannathan (1988), such as depositors' misinterpretation of liquidity withdrawals as informational withdrawals, can then be studied. Finally, the model in this paper can also be used to investigate the feasibility of mutual insurance among banks. As suggested by Calomiris (1990), banks can form a coalition to monitor and insure one another. Because banks are usually better informed than depositors, monitoring of banks by other banks may be more efficient. The mutual insurance arrangement gives banks a strong incentive to monitor one another. Moreover, once the coalition is formed, depositors care about the health of the coalition rather than that of individual banks. It is interesting to see whether mutual insurance among banks can provide an effective mechanism for disciplining banks.[28]

Appendix

Proof of Lemma 2

There are three groups of agents in this model: bankers, informed depositors, and uninformed depositors. Bankers obviously prefer the no-with-

[28] For more discussion on this proposal, see Calomiris (1990) and Chen (1994, chap. 2).

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drawal equilibrium since a bank's expected profit is positive in the no-withdrawal equilibrium and is zero in the panic run equilibrium. Note that a depositor's expected utility in the panic run equilibrium is

$$\frac{1}{y}u(y) + \left(1 - \frac{1}{y}\right)u(0),$$

which is less than $u(y)$. In the no-withdrawal equilibrium, the informed depositors' expected utility is strictly larger than $u(y)$ since they can always successfully withdraw. From the analysis in Section III, the uninformed depositors' expected utility in the no-withdrawal equilibrium is also larger than $u(y)$. Therefore, all parties prefer the no-withdrawal equilibrium to the panic run equilibrium. Q.E.D.

Proof of Proposition 1

To show (12) from (8), note that

$$\frac{a_1(K_1)}{1 - a_1(K_1)} = \frac{\eta_0 a_b^{K_1+1}(1 - a_b)^{N_1-K_1} + (1 - \eta_0) a_g^{K_1+1}(1 - a_g)^{N_1-K_1}}{\eta_0 a_b^{K_1}(1 - a_b)^{N_1-K_1+1} + (1 - \eta_0) a_g^{K_1}(1 - a_g)^{N_1-K_1+1}}.$$

Condition (8) can then be written as

$$\eta_0 a_b^{K_1}(1 - a_b)^{N_1-K_1+1}\left[\frac{a_b}{1 - a_b} - z(y, x)\right] \leq (1 - \eta_0) a_g^{K_1}(1 - a_g)^{N_1-K_1+1}\left[z(y, x) - \frac{a_g}{1 - a_g}\right].$$

Taking the log of both sides and rearranging the expression, we get (12). The result that K_1^* is decreasing in η_0 is obvious from (12). Other parts of the comparative statics can be proved using the facts that the right-hand side of (12) is increasing in $z(y, x)$, and $z(y, x)$ is increasing in x and is decreasing in y. Q.E.D.

Expressions for $\overline{U}_{u1}$, $\overline{U}_{u2}$, *and* $\overline{U}_i$

$$\begin{aligned}
\overline{U}_{u1} &\equiv a_0 u_b(y) + (1 - a_0) u_g(y, x),\\
\overline{U}_{u2} &\equiv \eta_0\{(1 - p_b^p)[a_b u_b(y) + (1 - a_b) u_g(y, x)] + p_b^p u_p(y)\}\\
&\quad + (1 - \eta_0)\{(1 - p_g^p)[a_g u_b(y) + (1 - a_g) u_g(y, x)] + p_g^p u_p(y)\},\\
\overline{U}_i &\equiv \frac{N_1}{N}[a_0 u(y) + (1 - a_0) u_g(y, x)] + \frac{N - N_1}{N}\\
&\quad \cdot (\eta_0\{(1 - p_b^p)[a_b u(y) + (1 - a_b) u_g(y, x)] + p_b^p u_p(y)\}\\
&\quad + (1 - \eta_0)\{(1 - p_g^p)[a_g u(y) + (1 - a_g) u_g(y, x)] + p_g^p u_p(y)\}),
\end{aligned}$$

where

$$u_p(y) \equiv \frac{1}{y} u(y) + \left(1 - \frac{1}{y}\right)u(0).$$

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Proof of Proposition 3

I first show that there exists a $\pi > 0$ such that parts *a*–*c* hold given this π. In the deposit insurance system, at date 1 a banker will try to speculate on the short-term project if and only if the outcome of the long-term project is adverse. The date 0 expected utility of a depositor who buys deposit insurance is

$$\bar{U}'_u \equiv \alpha u(y) + (1 - \alpha) u(x - \pi).$$

Given the proposed equilibrium, the expected utility of an informed depositor who does not buy deposit insurance is

$$\bar{U}'_i \equiv \alpha u(y) + (1 - \alpha)[a_0 u(y) + (1 - a_0) u(x)].$$

Given the belief that all the other uninformed buy deposit insurance, the expected utility of the uninformed depositor who does not buy deposit insurance is

$$\bar{U}_d \equiv \frac{N_1}{N}(\alpha u(y) + (1 - \alpha)\max\{u(y), a_0 u(0) + (1 - a_0) u(x)\}) + \frac{N - N_1}{N}\left(\alpha u(y) + (1 - \alpha)\sum_{i=0}^{N_1} \text{prob}(K_1 = i) \cdot \max\{u(y), a_1(i) u(0) + [1 - a_1(i)] u(x)\}\right),$$

where $\text{prob}(K_1 = i)$ is the probability that $K_1 = i$. The deviating uninformed depositor can use K_1 to decide whether to withdraw at date 1, so given K_1, her expected utility is

$$\alpha u(y) + (1 - \alpha)\max\{u(y), a_1(K_1) u(0) + [1 - a_1(K_1)] u(x)\}.$$

Note that $\bar{U}'_i > \max\{\bar{U}_u, \bar{U}_d\} > u(y)$, $\bar{U}'_i > \bar{U}_i$, $\bar{U}'_u > \bar{U}'_i$ when $\pi = 0$, and $\bar{U}'_u = u(y)$ when $\pi = x - y$. Therefore, we can find a $\pi \in (0, x - y)$ such that, given this π,

$$\bar{U}'_i > \bar{U}'_u > \max\{\bar{U}_u, \bar{U}_d\}.$$

The expression above implies that, given this π, ''all the uninformed buy deposit insurance and none of the informed do'' can be supported as an equilibrium. Moreover, since the uninformed depositors have more incentive to buy deposit insurance than the informed and a depositor's incentive to buy deposit insurance is decreasing in the number of depositors who buy deposit insurance, the proposed equilibrium is in fact the unique equilibrium. It is obvious that part *b* holds in this equilibrium. As to part *c*, bankers and the informed depositors are better off when the deposit insurance system is imposed. The uninformed depositors also prefer deposit insurance since $\bar{U}'_u > \bar{U}_u$. Therefore, part *c* also holds. To show that it is possible that the insurer's expected profit is positive, let π^* denote the fairly

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priced insurance premium. It can be verified that

$$\pi^* = a_0\left\{\bar{x} - \frac{1 - [\alpha + (1 - \alpha)\beta]y}{(1 - \alpha)(1 - \beta)}\right\}.$$

The insurer's expected profit is positive if

$$\bar{U}_i' > \bar{U}_u' > \max\{\bar{U}_u, \bar{U}_d\}$$

when $\pi = \pi^*$. This condition is satisfied for the numerical example discussed in Section IV. Therefore, the insurer's expected profit in the proposed deposit insurance system that satisfies parts *a*–*c* can be positive. Q.E.D.

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